

Task Completion Report

CMake AddLibrary.cmake Deduplication

Date: 2026-03-11 00:58:49

Summary

Eliminated duplicate AddLibrary.cmake files between the main project and the rznnode submodule while maintaining rznnode's ability to build independently.

Problem

- Two identical AddLibrary.cmake files existed:
 1. cmake/AddLibrary.cmake (main project)
 2. source/Core/rznnode/cmake/AddLibrary.cmake (submodule)
- rznnode is a git submodule that needs to remain independently buildable
- Duplicate code creates maintenance issues and potential drift

Solution

Implemented a thin wrapper pattern with configurable options:

1. Canonical Source (rznnode/cmake/AddLibrary.cmake):
 - Contains all implementation logic
 - Added configurable cache variables:
 - * RZNODE_CUDA_EXTRA_FLAGS: Extra CUDA compiler flags
 - * RZNODE_LINK_PYTHON_TO_NANOBIND: Python linking behavior
2. Thin Wrapper (cmake/AddLibrary.cmake):
 - Only 12 lines (down from 332)
 - Sets Ruzino-specific configuration
 - Delegates to rznnode's canonical version

Architecture

Main Project (Ruzino):

```
cmake/AddLibrary.cmake (wrapper)
|-- Set: RZNODE_CUDA_EXTRA_FLAGS="--forward-unknown-to-host-compiler;-c"
|-- Set: RZNODE_LINK_PYTHON_TO_NANOBIND=ON
+-- include --> source/Core/rznnode/cmake/AddLibrary.cmake
```

rznnode Standalone:

```
cmake/AddLibrary.cmake (canonical)
|-- Uses default values for all options
+-- Fully functional on its own
```

Files Changed

Task Completion Report

1. source/Core/rznnode/cmake/AddLibrary.cmake

- Added configurable cache variables (RZNODE_CUDA_EXTRA_FLAGS, RZNODE_LINK_PYTHON_TO_NANOBIND)
- Made CUDA extra flags configurable
- Made Python linking behavior configurable

2. cmake/AddLibrary.cmake

- Reduced from 332 lines to 12 lines
- Now a thin wrapper that includes rznnode's version
- Sets Ruzino-specific configuration before inclusion

Benefits

- No code duplication: Single source of truth in rznnode
- rznnode independent: Can be cloned and built standalone
- Backward compatible: Works in both source tree and installed scenarios
- Easy maintenance: Only edit rznnode/cmake/AddLibrary.cmake
- Flexible: Projects can customize behavior via cache variables
- Reduced line count: 332 -> 12 lines in main wrapper

Verification

- CMake configuration: PASSED
- Build target nodes_core: PASSED
- Both development and installation paths work correctly